

Profile

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Arpit Sharma

Experience Summary

Accomplished Senior Software Engineer with more than 9 years of experience, including 4 years at Siemens Advanta Technology, delivering high-performance applications using Modern C++ (C++11/14/17), OOP, STL, Smart pointers, Design pattern and data structures & algorithms. Proficient in the full SDLC, specializing in application development, microservices (gRPC, Protocol Buffers), and system architecture.

Skilled in Object-Oriented Design (OOD), System Design, TCP/IP socket programming, multithreading and understanding of SOLID principle. Adept at leveraging Visual Studio (2022, 2019, 2017) and Linux platforms for robust solutions. Experienced in CI/CD pipelines, utilizing tools like JIRA, GitLab, and JFrog Artifactory for seamless deployment.

Strong implementation knowledge of the Google Test Framework and static code analysis tools like SonarQube, ensuring high-quality, maintainable codebases. Committed to driving innovation and delivering scalable, secure software solutions.

Microsoft Azure Cloud Fundamentals Certified, with expertise in Azure cloud concepts, security, identity access, cost management, compliance, and governance. Proficient in OS and platform migrations, including Windows 7 to Windows 10 transitions, Visual Studio upgrades (VS 2010 to VS 2017), and 32-bit to 64-bit application migrations. Experienced in Windows OS customization, Microsoft security patches, and Nessus vulnerability scanning.

Professional Experience

Current Company Experience : Sr. Software Engineer at *Siemens Advanta technology (since Nov 2021 – till now)*.

Previous Company: 3.8 years of working experience with *General Electric Healthcare (GE Healthcare) (May 2018-Nov-2021)*.

Previous Experience : 1.10 years working experienced with *TAS Information Intelligence (Mar 2016- May 2018)*

Job Profile:

- **Software Development:** Extensive experience in developing core modules using C/C++ with strong proficiency in Object-Oriented Programming (OOP), Smart Pointers, Data Structures, STL library classes, Design Pattern and understanding of SOLID principle.
- **Networking & Multithreading:** understanding of TCP/IP socket programming and multithreaded application development in C++.
- **Microservices & Protocols:** Proficient in microservices architecture, including gRPC and Protocol Buffers, for building scalable systems.
- **Operating Systems:** In-depth knowledge of data structure programming in Linux and Windows environments, with hands-on experience in driver-side code development.
- **Cloud Expertise:** Certified in Microsoft Azure with comprehensive knowledge of cloud concepts, services, security, identity management, cost optimization, compliance, and governance.
- **Migration & Modernization :** Successfully led application migrations from 32-bit to 64-bit on Windows 10 and Visual Studio upgrades from 2010 to 2017.
- **Requirement Analysis & Testing:** Skilled in analyzing client requirements, designing test plans, creating test scenarios, and developing test cases for module, integration, and system testing.
- **Scripting & Automation:** Basic proficiency in Python, batch scripting, and CMakeList.txt configuration for build automation.

- **Static Analysis & Debugging** : Advanced expertise in static analysis tools like SonarQube and remote debugging using tools such as msvcmon.exe via Visual Studio.
- **Tracing & Analytics** : Proficient in using tracing tools like RTTraceViewer for log analysis and performance optimization.
- **Used Tracing tools** like RTTraceVlewer log analytics tool .

Certification :

- Got certified with **MicroSoft Azure Fundamental – AZ-900** .

AI Tool :

- Git-Hub Co-Pilot
- GPT

Technical Skills:

- **Programming Language** : C++, C , Core Python.
- **MicroService**: gRPC and Protocol Buffer
- **Development tools** : Visual Studio 2022, Visual Studio 2019, Visual Studio 2015, GCC, CMake, Make
- **Platforms** : Windows, Linux.
- **FrameWork/Libraries**: STL, Google Test Framework, BOOST
- **Databases**: LMDB, Oracle
- **Version Control** : GIT, GitLab, TFS(Azure devops)
- **Deployment tool** : Gitlab CI/CD, JFROG CaptainArtifactory
- **Other tools** : JIRA, , SonarCube, GDB, Valgrind, RTTraceViewer, Basic knowledge of docker and Kubernetes.
- **Scripting** : Python, Batch, bat, CMakelist.txt.
- **OS & Platform and Application Migration**: Application migration 32bit-64bit and Platform Windows 7 to Windows 10.
- **Methodolgies** : Agile Methodologies

Domain Knowledge :

- Siemens Advanta Technology – Industrial Automation and Discreate Domain.(2021-2023)
DIGI (Digitalization and Technology) (2023-till now)
- GE Healthcare (General Electric) - Healthcare Domain.
- TAS Intelligent Information – Data Mining Domain

Academic Qualification

- Pursued **Bachelor of Engineering** in 2014 from Computer Science and Engineering at VNS Institute of technology with an aggregate of **72.4%**. (RGPV UNIVERSITY BHOPAL).
- Passed with **72.6%** of marks in **10+2** education's in 2010 at Central Academy HR. Sec. School, Vidisha [M.P]
- Passed with **77.2%** of marks in **10th** education's in 2008 at Saket Garima HR. Sec. School, Vidisha. [M.P.]

Project Details

Project 1: Current Company Project : - EDI (BISRTE)

Description	EDI means Electronic Data Interchange (EDI) is the computer-to-computer exchange of business documents in a standard electronic format between business partners. It's a combination of Java, C++, SQL database, AWS Cloud. Mine responsibility is completely into the C++ side. BisRTE code is written into the C++. BISRTE stands for Business Integration Server Run Time Environment. In which different components are there for different jobs like File2ComDB, VVT (Pre-Dispatcher), BisKomin, BisIn, BisOut. So we can exchange the business document through these components.
Duration	March 2024 – Till date (running project)

Team size	16
Contribution	• Feature Development: Designed and implemented feature-rich code in C++ for EDI projects, ensuring alignment with functional and performance requirements. • Design and Requirement Analysis: Actively participated in reviewing and refining design documents and requirements to ensure clarity, feasibility, and adherence to project goals. • Unit Testing and Code Coverage: Developed and executed comprehensive unit tests using the Google Test framework, achieving high code coverage and ensuring robust software quality. • Code Optimization and Bug Fixing: Utilized static code analysis tools to identify and resolve bugs, improve code performance, and enhance maintainability, resulting in more efficient and reliable systems. • GitLab Maintainership: Owned and managed code deployment pipelines in GitLab, ensuring seamless integration, version control, and deployment processes for myself and whole development teams

Project Details

Project 2: Current Company Project : - HMI Runtime

Description	HMI means Human Machine Interface. It is a Panel which is communicated with PLC and the controller and provide the runtime environment . We created the project on TIA (Total Integration Automation tool) Using this TIA tool customer can create their own project and used it in the industrial area . For Ex – This can be used in automation industry , Sensor , discrete automation domain etc
Duration	Nov 2021 – March 2024
Team size	8
Contribution	• Project Contribution : Developed and maintained high-quality C++ code. • Feature Development : Done the implementation of Driver Code . • FrameWork: Implementation of Google test framework with C++. • Code Coverage: Implementation of Lcov Code coverage using Python scripting and CMakeList.txt file. • Bug Analysis : Analyze the code and debug the code using debugging tool (visual studio debugger) and logsto trace the bug's begin in the codebase. • Bug Fixing : Fixing the bug and Improve the code performance using static code analyzer . • Gitlab Ownership: Own the maintainership of Azure git code deployment

Project Details

Project 3: GE HealthCare Project : - Carescape Central Solution

Description	This Project is based on the Centralized ECG machine called Carescape Central Station (Healthcare Domain) . In this we have monitored the Heartbeat , Respiratory Rate , Spo2 rate Heart rate through the centralized Monitored using Simulator .
Duration	May 2018 – Nov 2021
Team size	25
Contribution	• Feature Development : Developed and maintained robust C++ code, contributing to the design, implementation, and optimization of software features. • Unit testing: Performed unit testing on key features using the Google Test framework, enhancing code reliability and identifying defects early in the development cycle. • OS and Platform Migration: Executed OS and platform migration, successfully transitioning applications from Windows 7 to Windows 10 and converting 32-bit applications to 64-bit architecture, ensuring compatibility and performance on modern

	systems. • Production Support: Provided critical development support for production environments, delivering emergency fixes with a focus on quality, reliability, and minimal downtime. • Project Planning: Collaborated on project planning and risk management, identifying potential challenges and proposing mitigation strategies to ensure timely delivery. • Debugging: Worked with the team to debug complex code issues, utilizing debugging tools (e.g., GDB, Visual Studio Debugger) to isolate and resolve defects efficiently.
Tools	DISM , WSIM , Rally , Defect Management tool , Clear Quest , SVN
Language / OS	C++ with STL Library , Multithreading , boost / Visual Studio (VC++) / Linux

Project Detail :

Project 4: TAS Information Intelligence Project :- TR Estimate

Description	This Project is based on the "DATA MINING " (domain= data mining). Language used in this Project is C++. In this we processes the file which takes the PDF Page as an input and parse the PDF pages and writes into the database and extract the those financial data which has been required according to client requirement and differentiate the that financial data based on the ever year, quarterly year , half yearly and finding the Delta on those data from the previous year and current year . In that we are using the C++ with STL library , Multithreading , Boost library and Basic Socket Proramming.
Duration	May 2016 – Mar 2018
Team size	5
Contribution	• Feature Involment : Involved in developing and reviewing design documents, requirement, code. • Feature Development : Feature implementation using STL library and add unit test cases for some features of code • Feature Development : Provided development support with emergency fixes on production environment with high level of commitment and quality. • Project Planning: Helping in project planning and risk management. • Team Collabrations & Debugging: Collaborated effectively with cross-functional teams to debug complex code, driving improvements and maintaining high standards of code quality.
Domain/Language/OS/Tools used	C++ with STL Library , Multithreading , boost / Visual Studio (VC++) / Linux

Project Details

Project 5: Self Project: - E-Commerce microservice application using gRPC and Protocol Buffer

Description	The application is split into loosely coupled microservices (Ex- user detail, inventory, payment , product, order and client), each responsible for a specific business function. These services communicate using gRPC , a high-performance, open-source RPC (Remote Procedure Call) framework developed by Google. Protocol Buffers is used to define the service interfaces and message structures in a language-agnostic, efficient, and extensible format. This approach ensures scalability, low-latency communication, and maintainability.
Duration	April 2025-June 2025
Team size	1
Contribution	Feature Development using microservices

	• Product Service: Manages product catalog (e.g., add, update, delete, or retrieve product details). • User Service: Handles user authentication, registration, and profile management. • Order Service: Processes customer orders, including creation, retrieval, and status updates. • Payment Service: Manages payment processing and transaction status. • Inventory Service: Tracks product stock levels and updates inventory.
Tools and Database	Visual Studio 2022, Oracle SQL
Language / OS	C++

Personal Information

- **Sex:** Male
- **Nationality :** Indian
- **DOB :** 23rd May 1992
- **Contact no:** +91 8050182171
- **Languages Known:** English , Hindi
- **Current Location:** 101 Apt. No. 2nd floor , Dreamz Namah Apartment ,Roopenagrahara , Near Silk Board , Bangalore (560068)

Declaration:

I hereby declare that the above-mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above mentioned particulars

Arpit Shama